Clinton Enwerem | Academic CV

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GitHub LinkedIn Google Scholar

Research Interests

Robot Learning, Dexterous Robotic Grasping & Manipulation, Multi-Modal Uncertainty Estimation.

Education

2021-Date **PhD, Electrical & Computer Engineering**, University of Maryland, College Park, MD, USA. Advisors: Professor John S. Baras and Professor Calin Belta. Relevant Coursework: *Decision-Making for Robotics, Network Control Systems, Decision Making Under Uncertainty, Random Processes, Advanced Digital Signal Processing, Nonlinear Control Systems, Optimal Control, System Theory, Convex Optimization, Formal Methods for Control & Dynamical Systems, Multimodal Foundational Models.*

2014-2018 **B.Eng., Electrical Engineering**, University of Nigeria, Nsukka, Enugu, Nigeria. Graduated with Highest Honors. Emphasis: Control Theory. *Completed coursework in engineering numerical analysis, control systems, electrical machines and drives, computer architecture, electronics, and communication systems.*

Selected Publications

Preprints/Articles In Review

2026 **Clinton Enwerem**, John S. Baras, and Calin Belta, "EquiDexFlow: Contact-Grounded SE(3)-Equivariant Dexterous Grasp Generative Flows," arXiv preprint, 2026. arXiv link

2026 **Clinton Enwerem**, Shreya Kalyanaraman, John S. Baras, and Calin Belta, "Variational Neural Belief Parameterizations for Robust Dexterous Grasping under Multimodal Uncertainty," arXiv preprint, 2026. Extended preprint with additional experiments, analysis, and discussion. arXiv link

2026 **Clinton Enwerem**, John S. Baras, and Calin Belta, "Risk-Constrained Belief-Space Optimization for Safe Control under Latent Uncertainty," arXiv preprint, 2026. arXiv link

2023 **Clinton Enwerem**, John S. Baras, and Danilo Romero, "Distributed Optimal Formation Control for an Uncertain Multiagent System in the Plane," arXiv:2301.05841 [cs, eess], Jan. 2023.

In Conference Proceedings

2025 **Clinton Enwerem**, Aniruddh G. Puranic, John S. Baras, and Calin Belta, Safety-Aware Reinforcement Learning for Control via Risk-Sensitive Value Iteration and Quantile Regression. In the proceedings of the 64th IEEE Conference on Decision and Control (CDC), 2025.

2024 **Clinton Enwerem**, Erfaun Noorani, John S. Baras, and Brian M. Sadler, Robust Stochastic Shortest-Path Planning via Risk-Sensitive Incremental Sampling, In the proceedings of the 63rd IEEE Conference on Decision and Control (CDC), 2024.

2024 **Clinton Enwerem** and John S Baras. Safe Collective Control under Noisy Inputs and Competing Constraints via Non-Smooth Barrier Functions. In the 2024 European Control Conference (ECC), pp. 3762–3768. IEEE, 2024.

2023 **Clinton Enwerem** and John S. Baras, "Consensus-Based Leader-Follower Formation Tracking for Control-Affine Nonlinear Multiagent Systems," in the 9th International Conference on Control, Decision and Information Technologies (CoDIT), Rome, Italy: IEEE, Jul. 2023, pp. 1226–1231. doi: 10.1109/CoDIT58514.2023.10284199.

Journal Articles

2024 **Clinton Enwerem** and John S. Baras, Formation Tracking for a Class of Uncertain Multiagent Systems: A Distributed Kalman Filtering Approach, IEEE Control Systems Letters, Volume 8, 2024.

Research Experience

Institute for Systems Research (ISR), University of Maryland

8/2021-Date **Graduate Research Assistant**

- Collaborate with PI, postdoctoral scholars, and graduate researchers to design novel robust grasping and manipulation algorithms for robotic arms equipped with anthropomorphic hands.
- Implement ROS 2-compliant software in Python and C++ for proposed grasp planning and control methods.
- Validate algorithms through large-scale simulation using high-fidelity simulators (MuJoCo, Drake) and RL sandboxes (OpenAI Gym, Safety Gymnasium).
- Author conference and journal papers, technical reports, and presentations to communicate research results.

Electrical Engineering Department, University of Nigeria, Enugu, Nigeria

9/2018-3/2021 **Research Assistant**

- Developed open-source software implementing Active Disturbance Rejection Control (ADRC) for robust motor control.
- Co-authored and published a journal paper reporting experimental and theoretical findings.

8-10/2017 **Undergraduate Research Assistant**

- Designed a feedback control algorithm for first-order-plus-dead-time processes; implementation and accompanying paper available online.

Professional Experience

Institute for Systems Research, College Park, MD

6-8/2023 **Summer Research Assistant**

- Formulated a multi-agent safety-critical control problem as a chance-constrained mathematical program. Supervisor: Professor John S. Baras.
- Proposed a novel solution based on Boolean-composed control barrier certificates.
- Wrote software to validate approach, and prepared a research paper to summarize results.

MATRIX Lab, USM at Southern Maryland, California, MD

6-8/2022 **Research Intern**

- Conducted system identification experiments to validate a quadrotor model. Supervisor: Dr. Danilo Romero.
- Developed a Lagrangian-based optimal swarm control algorithm for coordinating 10 Crazyflie quadrotors tasked with formation tracking under localization uncertainty.
- Wrote ROS-compliant and performant software (Python) implementing the control algorithm, and prepared a research paper and a technical report to summarize research findings.

Kognitive Robotics, Lagos, Nigeria

3/2020-2/2021 **Robotics Engineer**

- Engineered the complete power and sensor-actuator interface circuitry for a mobile robot from scratch, using Altium for schematic capture and PCB design.
- Supported robot hardware and ROS(1)-compliant software realization efforts for the company's MVP.

Robotics & Artificial Intelligence Nigeria (RAIN), Ibadan, Nigeria

3/2020-2/2021 **Robotics Trainee**

- Contributed to robotics and IoT projects across hardware and software stacks, encompassing computer-aided design, rapid prototyping, sensor fusion, control firmware development, and product testing.

Teaching Experience

ECE Department, University of Maryland

9/2025-12/2025 **Graduate Teaching Assistant**

- ENEE467 — Robotics Project Laboratory (with Prof. Calin Belta).
- Mentored 50+ undergraduate students in kinematics, 3D perception, motion planning, and control.
- Built reproducible Docker-based labs and testing pipelines spannig topics in manipulator kinematics, 3D perception, motion planning, robot control, and autonomous manipulation.
- Led weekly lab sessions and provided technical feedback on documentation and debugging.

2/2025-5/2025 **Graduate Teaching Fellow**

- ENEE661 — Nonlinear Control Systems (with Prof. John S. Baras).
- Led problem-solving sessions and discussions on nonlinear analysis and control, covering phase-plane and bifurcation analysis, and stability concepts including Lyapunov, input-to-state, L2, and L-infinity stability.

Technical Skills

Robotics	ROS 2, MuJoCo, Drake, GraspIt!.
RL	OpenAI Gym, Safety Gymnasium.
Robots/EoATs	UR3e/UR5, xArm7, LEAP Hand, RealHand L6, Robotiq Hand-E.
Tools	git, GitHub, GitLab, Docker.
Programming	Python, C++, Matlab, Bash.
Optimization	cvxpy, Gurobi, Mosek, Pyomo.
Perception/ML	PyTorch, OpenCV, TensorFlow.

Honors & Awards

2024	IEEE CSS Student Travel and Workshop Support: Conference travel award to attend CDC'24.
2022	Microsoft Diversity in Robotics & Autonomy PhD Fellowship (2022-2023), MRC & Microsoft.
2022	ROSCon Diversity Scholarship: Travel grant to attend ROSCon 2022 in Kyoto, Japan.
2021	Finalist, Engineers' League, Pan-African Robotics Competition, Rwanda.
2021	CIT Dean's Fellowship, Carnegie Mellon University, Africa Campus, Kigali, Rwanda.
2021	Dean's Fellowship, University of Maryland, College Park, MD, United States.
2020	Scholar, Stanford Exposure to Research & Graduate Education, Stanford University, CA, USA.
2020	EducationUSA Opportunity Funds Program Scholarship, U.S. Consulate General, Lagos, Nigeria.
2020	Sole Recipient (Nationwide), Door Foundation Leadlight Scholarship, RAIN.
2016-2018	Agbami Science & Technology Scholarship, Chevron: Merit-based undergraduate scholarship.
2015-2018	MTN Foundation Scholarship: Nationwide merit-based undergraduate scholarship.

Professional Training & Development

1/2020-2/2021 **Certificate in Robot Development & Automation, RAIN**

- Completed graduate-level coursework and projects in robotics, control theory, ML, and IoT.

Open Courseware

Fall 2025	Robot Learning (USC).
Summer 2025	Planning & Learning in Robotics (UCSD), Principles of Robot Autonomy I & II (Stanford).
Summer 2023	Bayesian Statistics (UCSD, Coursera).
Summer 2022	Autonomous Navigation for Flying Robots (TUM, edX).

Professional Service, Outreach, & Mentoring

2023-2026	Peer Reviewer: Heliyon, MED'23, ECC'24, ACC'25, L4DC'25, CDC'25, CDC'26, Elsevier (Robotics and Autonomous Systems).
2021-Date	Member, Black in Robotics (BiR): U.S.-based organization promoting Black representation in robotics.
2021-2022	Mentor, EducationUSA & iScholar Initiative: Guided STEM graduates from underrepresented backgrounds to secure fully-funded graduate admissions at Stanford, Purdue, and Penn State.

Miscellany

Languages	English (Fluent; TOEFL iBT: 110/120), Japanese (Conversational).